

Tunisia, audited — investor economics

The FX bridge by leg, per-pathway DSCR computed in code, the capital stack, and the exact signatures that move “commercially validated” from zero.

Provenance law: MEASURED = cited anchor · DERIVED = arithmetic on measured values · ASSUMED = declared default. commercially-validated = 0 — no buyer has signed anything; published as zero on purpose. Generated 2026-09-09 by nations_audit.py / make_nations_pdfs.py — the same code that renders the page.

1. The FX bridge, tiered by confidence

LEG	\$M/YR	GRADE
Strategic metals (U + heavy REE, acid stream)	\$75-86M	STRONG
Solar fuel substitution (500 MW)	\$95-116M	PLAUSIBLE
Hydrogen from solar surplus	\$21-25M	<i>SPECULATIVE</i>
Terfas (desert truffles)	\$14-29M	PLAUSIBLE
Food import substitution	\$30-30M	<i>SPECULATIVE</i>
Compute colocation services	\$15-15M	<i>SPECULATIVE</i>
ESG financing delta	\$8-12M	PLAUSIBLE
Total (mid)	~\$285M/yr	printed \$312M → corrected

\$80M carries a proven flowsheet or a cited anchor; \$137M is gated; \$68M rests on assumed volume or price. Zero dollars are commercially validated — the first buyer moves the ledger.

2. Pathway economics — computed in code (nations_v5.py)

P2 · Strategic metals (acid-stream U + heavy REE)

LINE	VALUE
Revenue	\$75-86M/yr (716.5k lbs U&sb3;O&sb8; @ \$80-95/lb + ~585 t REO)
OPEX	\$54-74M/yr (SX \$59-83/lb + shared circuit)
Gross margin	0-37%
CAPEX / payback	~\$100M / 6.3 yr (mid)
DSCR (60% debt @7%/15y)	2.40 — passes at contract prices
Gate	pilot U recovery ≥85% · all-in ≤\$75/lb · offtake signed

P3 · Solar IPP (500 MW) — the honest finding

LINE	VALUE
Revenue / OPEX	\$27.2M / \$8M (500 MW × 20% CF × \$31/MWh)
CAPEX	~\$550M
DSCR @8%/20y	0.46 — fails
DSCR @4.5% DFI/25y	0.69 — fails
Bankable region (DSCR ≥1.15)	\$0.70/W + 21% CF · \$0.70/W + 22% CF · \$0.70/W + 23% CF · \$0.70/W + 24% CF · \$0.80/W + 24% CF
Clearing PPA	≥\$45652/MWh at mid-case capex/CF
Verdict	bankable ONLY lean, inside the region, or with grant/sovereign support — published, not hidden

P4 • Municipal desalination (150,000 m³/d)

LINE	VALUE
Revenue	\$30.1M @\$0.55 → \$38.3M @\$0.70 blended
OPEX / CAPEX	\$24.6M (@\$0.45/m ³) / ~\$165M
DSCR @\$0.55	0.54 — fails
DSCR @\$0.70	1.36 — clears
Gate	≥60% contracted offtake · blended ≥\$0.68/m ³ · brine permit

3. The capital stack and the ask

Phase o: \$150-250k, 90 days, no construction — the price of replacing every projection with a measurement (itemized in the complete audit). **Then, module by module, on public gates:** \$58k health pilot · \$100M metals · \$550M solar · \$165M desal · \$5-30M truffles · \$2-5M food. Security per module: offtake assignment, STEG PPA, municipal contracts, MIGA/DFI channel — modeled, not committed.

What moves “commercially validated” from zero

GCT MOU + first assay batch · STEG PPA award · desal LOI ≥60% at ≥\$0.68/m³ · first truffle fruit + ≥€20/kg contract · 100-point water panel + 90-day health KPIs · 3,000-family cohort baseline. Each is a Phase-o/1 deliverable with a date.

What this document is: a concept-level engineering design whose numbers are computed in code from sourced primitives — recomputed, corrected, and applied into the design. Free and public.

What it is not: a commissioned state program, a bank-grade feasibility study, an investment offer, or a claim that any of this has been built. Nothing here is advice to any government or any investor. Nothing replaces geological, radiological, geotechnical or EPC-level surveys. Verify every figure and citation before relying on any of it.

Works cited

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6. [Rw9] 24/7 firming: 3.03 GW @20% CF = 0.606 GW avg (zero margin); 9.15 GWh/night; BESS round-trip 85% → 12-15 GWh; \$110-125/kWh → ~\$1.5B; solar+storage LCOE \$100-130/MWh — <https://www.energy-storage.news/battery-storage-system-prices-continue-to-fall-sharply-bnef-and-ember-reports-...>
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